Data Flow Diagram (DFD)

Project Name: Human Development Index (HDI) Prediction Using Machine Learning

External Entity

User (Government Official / Researcher / Student / Policy Maker)

The user enters the required development indicators into the system.

Process 1: Input Data Collection

Description:

- Accept user input.
- Collect:
  - Life Expectancy
  - Mean Years of Schooling
  - Expected Years of Schooling
  - GNI Per Capita

Output:
Validated input data.

Data Store 1: HDI Dataset

Description:
Stores the historical Human Development Index dataset used for model training and reference.

Contents:

- Life Expectancy
- Mean Years of Schooling
- Expected Years of Schooling
- GNI Per Capita
- HDI Category

Process 2: Data Preprocessing

Description:

- Check missing values.
- Clean the data.
- Encode categorical values (if required).
- Prepare the input for the machine learning model.

Output:
Preprocessed data ready for prediction.

Process 3: Machine Learning Prediction

Description:

- Load the trained Machine Learning model.
- Predict the HDI category based on user inputs.

Output:
Predicted HDI Category.

Data Store 2: Trained Model

Description:
Stores the trained Machine Learning model (model.pkl) used to make predictions.

Process 4: Display Result

Description:
Display the predicted HDI category to the user.

Possible Outputs:

- Very High HDI
- High HDI
- Medium HDI
- Low HDI

Data Flow

User

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│ Enter HDI Indicators

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Process 1

Input Data Collection

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Process 2

Data Preprocessing

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Data Store

HDI Dataset

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Process 3

Machine Learning Prediction

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│ Uses

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Data Store

Trained Model (model.pkl)

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Process 4

Display Prediction

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User

DFD Level-0

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|      User       |
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        |
        | Input:
        | Life Expectancy
        | Mean Years Schooling
        | Expected Years Schooling
        | GNI Per Capita
        v
+-------------------------------+
|  HDI Prediction System       |
|  - Validate Input           |
|  - Preprocess Data          |
|  - Predict HDI Category      |
|  - Display Result           |
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        |
        v
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| Predicted HDI Result |
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DFD Level-1

User

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(1) Input Data Collection

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(2) Data Preprocessing

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HDI Dataset

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(3) Machine Learning Model

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Trained Model (model.pkl)

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(4) Display Prediction

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User

Conclusion

The Data Flow Diagram illustrates how user input moves through the HDI Prediction System. The application collects socio-economic indicators, preprocesses the data, uses the trained machine learning model to predict the Human Development Index category, and returns the result to the user. This structured flow ensures accurate, efficient, and user-friendly HDI prediction.